

When Sarah Nelmes, a milkmaid, visited his practice with cowpox in 1796, Jenner decided to put his theory to the test. On May 14, he transferred blister fluid from Nelmes to both arms of James Phipps, the 8-year-old son of his gardener. Although Phipps became slightly ill over the next 9 days, by the tenth, he was well again. To find out if his experiment had worked, on July 1, Jenner deliberately infected Phipps with smallpox. To Jenner's relief, the boy not only survived, but further tests carried out throughout his long life proved he had developed immunity to smallpox.

Critical reception

Jenner published the full results of his study, which included vaccinations of 23 patients—including his own son—in 1798. Initially, his report was met with opposition and even ridicule among an uneducated public. Although the medical community had doubts about the ethics of his experimental method, Jenner's thorough, scientific approach was lauded. Over time, his procedures were accepted and improved and were used in America, the rest of Europe, and across the world. The rate of smallpox

BENJAMIN JESTY

Though his contribution only became clear following Jenner's more scientific achievements, Jesty was eventually recognized as one of the first successful smallpox vaccinators.



Jesty (1736–1816) was a farmer in Dorset, England, who had contracted cowpox due to his work with livestock. When a smallpox epidemic broke out in 1774, he took the unprecedented step of using a darning needle to transmit cowpox pus to scratches on the arms of his wife and two eldest sons. His sons suffered mild, local reactions, but his wife became gravely ill—although all three recovered fully. Initially mocked for his actions, Jesty was vindicated when his sons proved to be immune upon later exposure to the disease. His pioneering efforts came to light decades later, and in 1805, he was honored by the Original Vaccine Pock Institution in London.

mortality declined dramatically, and Jenner was recognized internationally for his work.

In 1853, three decades after his death, Jenner's findings finally convinced the British government to replace inoculation with compulsory vaccination. The smallpox vaccine has since saved several billion lives and triggered a global initiative in the 1970s that was the first, and still only, program to have eradicated an infectious disease.

**SMALLPOX CASES
HAD A MORTALITY
RATE OF 80%
IN THE 18TH
CENTURY** 

**TESTED EARLY
VACCINATION ON HIS
11-MONTH-
OLD SON**

**VACCINATIONS FOR
SMALLPOX MADE
COMPULSORY
IN ENGLAND
30 YEARS
AFTER HIS DEATH**